

Signals and Systems

Previous Year Questions • 1996–2026

GATE INSTRUMENTATION ENGINEERING

(IN)

Himanshu Agarwal & Anish Saha

PrepFusion™

Where Concept Meets Clarity!

Preface

Welcome to PrepFusion™!

This book is a complete compilation of the questions GATE has actually asked in Signals and Systems (IN), gathered from the original papers and arranged unit by unit and year by year. Nothing is paraphrased or reconstructed: every question appears as it was set, tagged with its year, session and marks.

Each question carries an identifier that maps one-to-one onto its worked solution in the companion solutions volume, so you can move between practice and explanation without a lookup table. Answer keys close every section, and answers revised by the examining authority (MTA) are marked where they occur.

Used end to end, the book shows you what the paper rewards — which topics repeat, how the framing has shifted over the years, and where your preparation still has gaps.

Meet your Authors

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- AIR 27 — GATE (ECE)
- AIR 45 — GATE (IN)
- 5+ years of teaching experience



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- AIR 66 — GATE (EE)
- 3+ years in teaching & the VLSI industry



> *"We built this book the way every aspirant deserves one to be built — organized strictly by unit and by year, so that every hour of practice maps directly onto how GATE actually tests you."*
>
> — Himanshu Agarwal & Anish Saha

How to Read the Question Numbers

Every question in this book carries a three-part identifier instead of a plain serial number:

Q3.24.7

3	Unit (chapter) number — Unit III of this book
24	GATE exam year — 2024 (two digits; 98 = 1998)
7	Question number within that unit and year

So **Q3.24.7** is the 7th question that GATE 2024 contributed to Unit III. Numbers are therefore not sequential through the book — they tell you the unit and the exam year at a glance. The same identifier is used in the questions book, the solutions book and the answer keys, so you can jump between them without a lookup table. In the PDF bookmarks panel, expand a unit to see every question ID and click one to go straight to it.

What the Bubbles Mean

Every question ends with three empty circles for you to shade in yourself:



 Easy  Medium  Hard

The companion solutions volume marks the same question with *our* recommended difficulty, using the identical bubbles. If you think a rating should be different, tell us directly — report it under the **Study Hub** section at **pyq.prepfusion.in**.

How to Read the Unit Snapshot

Every unit opens with a **Unit Snapshot** box summarizing that unit's real question data:

Questions	the real total for this unit.
MCQ, MSQ, NAT	that same total, split by question type.
1M, 2M	how many of those questions carry 1 or 2 marks.
Descriptive	questions worth more than 2 marks, or whose marks value isn't recorded.
Avg Weightage	the average marks this unit has contributed per real GATE sitting, not a raw year-by-year total — a year GATE ran multiple real sessions (a genuine forenoon/afternoon split, not the same paper's Set A/B/C/D booklet variants) is divided across those sessions first, so a heavily multi-session year cannot silently outweigh a single-session one. The ~ marks it as a rounded figure, not an exact one.
Range	the lowest and highest that same per-sitting value has been, with the year each occurred.

Avg Weightage and Range are computed from **2010 onward** only: GATE dropped from a 150-mark to a 100-mark paper in 2009, but that year itself was a one-off transition (60 questions, not the 65-question format that became standard from 2010 on) — so 2010 is the first year fully comparable to modern papers, and marks-based figures from earlier years are left out of these two figures specifically (every other count on this page still includes every year). They also depend on knowing which GATE years ran more than one real sitting for a given branch — our own research into this is thorough for some branches and still a work in progress for others. Treat both figures as a **beta**, best-effort estimate: directionally useful, not exact.

MSQ questions specifically weren't introduced until 2021 — an older-year-heavy unit showing **MSQ: 0** reflects that real exam history, not a data gap.

Worked example

Unit Snapshot*

Stream: EC **Questions:** 45 **MCQ:** 30 **MSQ:** 3 **NAT:** 12
1M: 20 **2M:** 22 **Descriptive:** 3 **Avg Weightage** ~4M **Range:** 2M(2015)–6M(2021)

45 real questions total; on average this unit has been worth about 4 marks per real GATE sitting since 2010, ranging from as low as 2 (in 2015) to as high as 6 (in 2021).

A quick way to use this: a high **Avg Weightage** points to a consistently high-yield topic worth prioritizing; a wide **Range** usually means the topic's importance has shifted over the years, so it's worth checking *which* years drove that before deciding how much weight to give it today.

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Scan any code with your phone camera, or tap the link beneath it. **pyq.prepfusion.in** is also where you can read this book's worked solutions online — and in its **Study Hub** section, report an error in any question or request a video solution for it.

Contributors

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PrepFusion

UNIT

1

Basics of Signals

Topics

1. Basics of Signals and Mathematical Foundation
2. Step and Rectangular Signal
3. Ramp and Triangular Signal
4. Impulse Signal
5. Miscellaneous Signals
6. Complex Exponential Signal
7. Periodicity of a Signal
8. Orthogonality of Signals
9. Energy, Average and Power of a Signal
10. Even, Odd, Conjugate Symmetry Signals
11. Discrete-Time Signals and Their Types
12. Classification of Discrete-Time Signals

Unit Snapshot*

Stream: IN	**Questions:** 28	**MCQ:** 20	**MSQ:** 0	**NAT:** 8
1M: 20	**2M:** 8	**Descriptive:** 0	**Avg Weightage** ~2M	**Range:** 1M(2025)–5M(2026)

**See preface for how Avg Weightage and Range are calculated.*

Q1.26.1 NAT 2026 2M

Let

$$f(t) = \begin{cases} 1, & t \in [0, 2] \\ -t + 3, & t \in [2, 3] \\ 0, & \text{otherwise} \end{cases}$$

Then $\int_{-\infty}^{\infty} f(\tau)\, d\tau =$ _____(rounded off to one decimal place)

Q1.26.2 MCQ 2026 2M

A signal $x(t)$ is shown below.

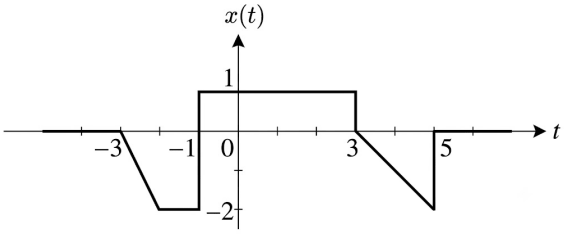
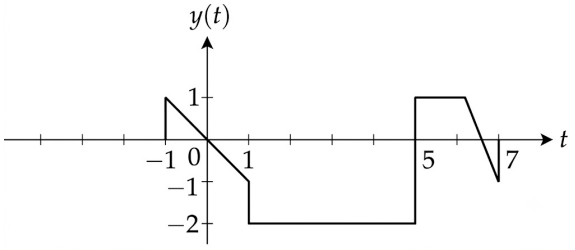


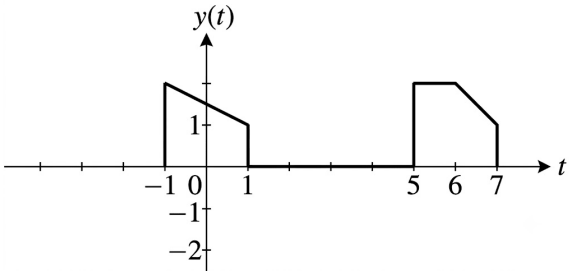
Fig. Q1.26.2

The plot of $y(t) = 1 - x(4 - t)$ is shown in

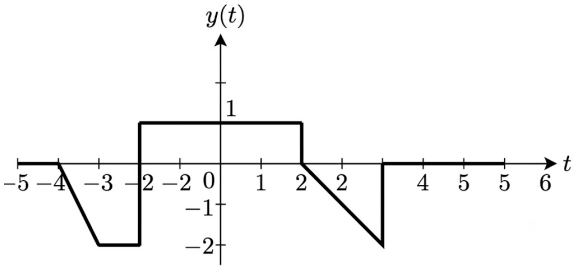
A.



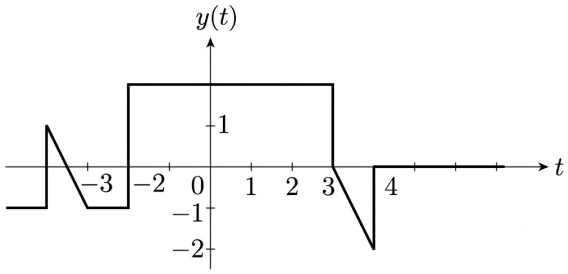
B.



C.



D.



Q1.26.3 MCQ 2026 1M

Given $x(t) = 4\sin(15\pi t) + 7\cos(4\pi t)$, where t is in seconds. The fundamental period of $x(t)$ is

A. 15 **B.** 60 **C.** 30 **D.** 2

Q1.25.1 MCQ 2025 1M

Let a continuous-time signal be

$$x(t) = e^{j9t} + e^{j5t},$$

where $j = \sqrt{-1}$ and t is in seconds. The fundamental period of magnitude of $x(t)$, in seconds, is

A. π **B.** $\frac{\pi}{2}$ **C.** $\frac{\pi}{5}$ **D.** $\frac{\pi}{9}$

Q1.23.1 NAT 2023 1M

Consider the discrete-time signal

$$x[n] = u[5 - n] - u[n + 3],$$

where the unit step is defined by

$$u[n] = \begin{cases} 1, & n \geq 0, \\ 0, & n < 0. \end{cases}$$

The smallest integer n for which $x[n] = 0$ is _____.

Q1.23.2 MCQ 2023 1M

A system has the transfer function

$$\frac{Y(s)}{X(s)} = \frac{s - \pi}{s + \pi}.$$

Let $u(t)$ be the unit step function. The input $x(t)$ that results in a steady-state output

$$y(t) = \sin \pi t$$

is

(A) $x(t) = \sin(\pi t)\, u(t)$

(B) $x(t) = \sin\left(\pi t + \frac{\pi}{2}\right) u(t)$

(C) $x(t) = \sin\left(\pi t - \frac{\pi}{2}\right) u(t)$

(D) $x(t) = \cos\left(\pi t + \frac{\pi}{4}\right) u(t)$

Q1.23.3 NAT 2023 1M

Let

$$y(t) = x(4t),$$

where $x(t)$ is a continuous-time periodic signal with fundamental period of 100 s. The fundamental period of $y(t)$ is _____s. (rounded off to the nearest integer)

Q1.21.1 MCQ 2021 1M

The signal $\sin(\sqrt{2\pi t})$ is

A. Periodic with period $T = 4\pi^2$

B. Periodic with period $T = \sqrt{2\pi}$

C. Not periodic

D. Periodic with period $T = 2\pi$

Q1.20.1 MCQ 2020 1M

Consider the signal

$$x[n] = \sin(2\pi n)u[n], \qquad u[n] = \begin{cases} 1, & n = 0, 1, 2, 3 \\ 0, & \text{otherwise} \end{cases}$$

The period of signal $x[n]$ is

A. 4 **B.** 1 **C.** 2 **D.** 3

Q1.20.2 NAT 2020 1M

Let

$$g[n] = \begin{cases} 1, & n = 0 \\ 0, & n = \pm 1, \pm 2, \pm 3, \ldots \end{cases}$$

and

$$h[n] = \begin{cases} 1, & n = 0, 3, 6, 9, \ldots \\ 0, & \text{otherwise.} \end{cases}$$

Consider

$$y[n] = h[n] \otimes g[n],$$

where $\otimes$ denotes the convolution operator. The value of $y[2]$ is _____.

Q1.19.1 NAT 2019 2M

A discrete-time signal

$$x[n] = e^{j\left(\frac{5\pi}{3}\right)n} + e^{j\left(\frac{\pi}{4}\right)n}$$

is down-sampled to the signal $x_d[n]$ such that $x_d[n] = x[4n]$. The fundamental period of the down-sampled signal $x_d[n]$ is _____.

Q1.18.1 MCQ 2018 1M

Consider signal

$$x(t) = \begin{cases} 1, & |t| \le 2 \\ 0, & |t| > 2 \end{cases}$$

Let $\delta(t)$ denote the unit impulse (Dirac-delta) function. The value of the integral $\int_0^5 2x(t-3)\delta(t-4)\,dt$ is

A. 2 **B.** 1 **C.** 0 **D.** 3

Q1.18.2 MCQ 2018 1M

Two periodic signals $x(t)$ and $y(t)$ have the same fundamental period of 3 seconds. Consider the signal

$$z(t) = x(-t) + y(2t + 1).$$

The fundamental period of $z(t)$ in seconds is

A. 1 **B.** 1.5 **C.** 2 **D.** 3

Q1.17.1 NAT 2017 1M

A periodic signal $x(t)$ is shown in the figure. The fundamental frequency of the signal $x(t)$ in Hz is

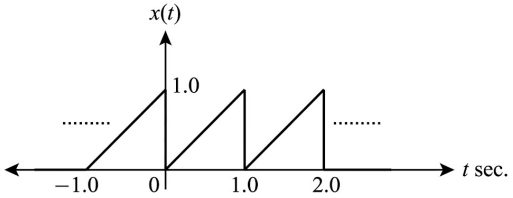


Fig. Q1.17.1

Q1.16.1 NAT 2016 2M

The fundamental period N_0 of the discrete-time sinusoid

$$x[n] = \sin\left(\frac{301}{4}\pi n\right)$$

is _____.

Q1.15.1 NAT 2015 2M

The fundamental period of the signal

$$x(t) = 2\cos\left(\frac{2\pi}{3}t\right) + \cos(\pi t),$$

in seconds, is _____ s.

Q1.13.1 MCQ 2013 1M

For a periodic signal

$$v(t) = 30\sin 100t + 10\cos 300t + 6\sin\left(500t + \frac{\pi}{4}\right)$$

The fundamental frequency in rad/s is

A. 100 **B.** 300 **C.** 500 **D.** 1500

Q1.11.1 MCQ 2011 1M

The continuous signal $x(t) = \sin\omega_0 t$ is a periodic signal. However, for its discrete-time counterpart $x(n) = \sin\omega_0 n$ to be periodic, the necessary condition is

A. $0 < \omega_0 < 2\pi$

B. $\frac{2\pi}{\omega_0}$ to be an integer

C. $\frac{2\pi}{\omega_0}$ to be a ratio of integers

D. None of the above

Q1.11.2 MCQ 2011 1M

The integral

$$\frac{1}{\sqrt{2\pi}}\int_{-\infty}^{\infty} t^2 e^{-t^2/2}\,\delta(1-2t)\,dt$$

is equal to

A. $\frac{1}{8\sqrt{2\pi}}e^{-1/8}$ **B.** $\frac{1}{4\sqrt{2\pi}}e^{-1/8}$

C. $\frac{1}{\sqrt{2\pi}}e^{-1/2}$ **D.** 1

Q1.10.1 MCQ 2010 1M

The integral

$$\int_{-\infty}^{\infty}\delta\left(t-\frac{\pi}{6}\right)6\sin(t)\,dt$$

evaluates to

A. 6 **B.** 3 **C.** 1.5 **D.** 0

Q1.09.1 MCQ 2009 1M

The fundamental period of $x(t) = 2\sin 2\pi t + 3\sin 3\pi t$ with t expressed in seconds, is

A. 1 sec **B.** 0.67 sec **C.** 2 sec **D.** 3 sec

Q1.08.1 MCQ 2008 1M

The fundamental period of the discrete-time signal $x[n] = e^{j\left(\frac{5\pi}{6}\right)n}$ is

A. $\frac{6}{5\pi}$ **B.** $\frac{12}{5}$ **C.** 6 **D.** 12

Q1.07.1 MCQ 2007 2M

Consider the discrete time signal,

$$x(n) = \left(\frac{1}{3}\right)^n u(n),$$

where

$$u(n) = \begin{cases} 1, & n \geq 0 \\ 0, & n < 0 \end{cases}$$

A signal $y(n)$ is defined as

$$y(n) = x(-n), -\infty < n < \infty.$$

Then $\sum_{n=-\infty}^{\infty} y(n)$ equals

A. $-\frac{2}{3}$ **B.** $\frac{2}{3}$ **C.** $\frac{3}{2}$ **D.** 3

Q1.07.2 MCQ 2007 1M

Consider the periodic signal $x(t) = (1+0.5\cos 40\pi t)\cos 200\pi t$ where t is in seconds. Its fundamental frequency, in Hz, is

A. 20 **B.** 40 **C.** 100 **D.** 200

Q1.06.1 MCQ 2006 2M

The Fourier series for a periodic signal is given as $x(t) = \cos(1.2\pi t) + \cos(2\pi t) + \cos(2.8\pi t)$. The fundamental frequency of the signal is

A. 0.2 Hz **B.** 0.6 Hz **C.** 1.0 Hz **D.** 1.4 Hz

Q1.06.2 MCQ 2006 1M

The root-mean-square value of a voltage waveform consisting of a superposition of 2 V dc and a 4 V peak-to-peak square wave is

A. 2 V **B.** $\sqrt{6}$ V **C.** $\sqrt{8}$ V **D.** $\sqrt{12}$ V

Q1.05.1 MCQ 2005 2M

The fundamental period of the sequence $x[n] = 3\sin(1.3\pi n + 0.5\pi) + 5\sin(1.2\pi n)$ is

A. 20 **B.** $\frac{2\pi}{1.3\pi}$ **C.** $\frac{2\pi}{1.2\pi}$ **D.** 10

Q1.01.1 MCQ 2001 1M

Which one of the following sequences is NOT a power signal?

A. Unit step sequence

B. $e^{j\omega_0 n}$

C. A periodic sequence

D. Unit ramp sequence

Answer Key

Note: * descriptive/evidence-based, no single definitive answer.

Q1.26.1 2.5	Q1.26.2 *	Q1.26.3 D
Q1.25.1 B	Q1.23.1 -3	Q1.23.2 C
Q1.23.3 25	Q1.21.1 C	Q1.20.1 B
Q1.20.2 0	Q1.19.1 6	Q1.18.1 A
Q1.18.2 D	Q1.17.1 1	Q1.16.1 8
Q1.15.1 6	Q1.13.1 A	Q1.11.1 C
Q1.11.2 A	Q1.10.1 B	Q1.09.1 C
Q1.08.1 D	Q1.07.1 C	Q1.07.2 A
Q1.06.1 A	Q1.06.2 C	Q1.05.1 A
Q1.01.1 D		

UNIT

II

Basics of Systems

Topics

1. Continuous-Time Convolution
2. Discrete-Time Convolution
3. Linear and Non-Linear Systems
4. Time Variant and Time-Invariant Systems
5. LTI Systems - Definition and Properties
6. Causal and Non-Causal Systems
7. Static and Dynamic Systems
8. Stable and Unstable Systems
9. Invertible and Non-Invertible Systems
10. Systems Involving Differential Equations

Unit Snapshot*

Stream: IN	**Questions:** 14	**MCQ:** 13	**MSQ:** 1	**NAT:** 0
1M: 7	**2M:** 7	**Descriptive:** 0	**Avg Weightage** ~2M	**Range:** 1M(2022)–4M(2025)

**See preface for how Avg Weightage and Range are calculated.*

Q2.26.1 MCQ 2026 2M

In the table shown below, List-I are the system responses to input $x(t)$ and List-II are the system properties. The correct matching is

List I		List II	
(M)	$x^2(t)$	(I)	Linear and Time-invariant
(N)	$x(-t)$	(II)	Non-linear and Time-varying
(O)	$\lvert x(3-t)\rvert$	(III)	Linear and Time-varying
(P)	$x(t-3)$	(IV)	Non-linear and Time-invariant

Fig. Q2.26.1

A. M-IV, N-III, O-II, P-I **B.** M-II, N-I, O-IV, P-III
C. M-III, N-IV, O-II, P-I **D.** M-I, N-IV, O-III, P-II

Q2.25.1 MSQ 2025 2M

Choose the Eigen function(s) of stable linear time-invariant continuous-time systems from the following options.

A. $e^{j\frac{2\pi}{3}t}$ **B.** $\cos\left(\frac{2\pi}{3}t\right)$ **C.** 2^t **D.** $\sin\left(\frac{2\pi}{3}t\right)$

Q2.25.2 MCQ 2025 2M

Let

$$y[n] = \frac{1}{\alpha}y[n-1] + x[n],$$

where $\alpha > 1$ and real, represent a difference equation of a causal discrete-time linear time-invariant system. The system is initially at rest.

If $x[n] = \delta[n-p]$, where $p > 10$, the value of $y[p+1]$ is

A. 0 **B.** 1 **C.** $\frac{1}{\alpha}$ **D.** $\frac{1}{\alpha^2}$

Q2.22.1 MCQ 2022 1M

The output of a system $y(t)$ is related to its input $x(t)$ according to the relation $y(t) = x(t)\sin(2\pi t)$. This system is

A. Non-linear and time-variant

B. Non-linear and time-invariant

C. Linear and time-invariant

D. Linear and time-variant

Q2.19.1 MCQ 2019 1M

The input $x[n]$ and output $y[n]$ of a discrete-time system are related as

$$y[n] = \alpha y[n-1] + x[n].$$

The condition on α for which the system is Bounded-Input Bounded-Output (BIBO) stable is

A. $|\alpha| < 1$ **B.** $|\alpha| = 1$ **C.** $|\alpha| > 1$ **D.** $|\alpha| < \frac{3}{2}$

Q2.14.1 MCQ 2014 1M

The impulse response of an LTI system is given as,

$$h[n] = \begin{cases} \frac{\omega_c}{\pi}, & n = 0 \\ \frac{\sin(\omega_c n)}{\pi n}, & n \neq 0 \end{cases}$$

It represents an ideal

A. Non-causal, low-pass filter

B. Causal, low-pass filter

C. Non-causal, high-pass filter

D. Causal, high-pass filter

Q2.13.1 MCQ 2013 1M

Which one of the following statements is NOT TRUE for a continuous time causal and stable LTI system?

A. All the poles of the system must lie on the left side of the $j\omega$ axis.

B. Zeros of the system can lie anywhere in the s-plane.

C. All the poles must lie within $|s| = 1$.

D. All the roots of the characteristic equation must be located on the left side of the $j\omega$ axis.

Q2.13.2 MCQ 2013 2M

The impulse response of a continuous time system is given by

$$h(t) = \delta(t-1) + \delta(t-3)$$

The value of the step response at $t = 2$ is

A. 0 **B.** 1 **C.** 2 **D.** 3

Q2.12.1 MCQ 2012 2M

The input $x(t)$ and output $y(t)$ of a system are related as

$$y(t) = \int_{-\infty}^{t} x(\tau)\cos(3\tau)\,d\tau.$$

The system is

A. time-invariant and stable

B. stable and not time-invariant

C. time-invariant and not stable

D. not time-invariant and not stable

Q2.11.1 MCQ 2011 1M

Consider a system with input $x(t)$ and output $y(t)$ related as follows

$$y(t) = \frac{d}{dt}\left\{e^{-t}x(t)\right\}$$

Which one of the following statements is TRUE?

A. The system is non linear

B. The system is time-invariant

C. The system is stable

D. The system has memory

Q2.10.1 MCQ 2010 2M

The input $x(t)$ and the corresponding output $y(t)$ of a system are related by

$$y(t) = \int_{-\infty}^{5t} x(\tau)\,d\tau.$$

The system is

A. time invariant and causal

B. time invariant and non-causal

C. time variant and non-causal

D. time variant and causal

Q2.09.1 MCQ 2009 2M

For input $x(t)$, an ideal impulse sampling system produces the output

$$y(t) = \sum_{k=-\infty}^{\infty} x(kT)\,\delta(t - kT)$$

where $\delta(t)$ is the Dirac delta function. The system is

A. non-linear and time-invariant

B. non-linear and time-varying

C. linear and time-invariant

D. linear and time-varying

Q2.08.1 MCQ 2008 1M

Which one of the following discrete-time systems is time-invariant?

A. $y[n] = n\,x[n]$ **B.** $y[n] = x[3n]$

C. $y[n] = x[-n]$ **D.** $y[n] = x[n-3]$

Q2.01.1 MCQ 2001 1M

The discrete LTI system with the following impulse response is non-causal.

A. $a^n u(n-2)$ **B.** $a^{n-2}u(n)$

C. $a^{n+2}u(n)$ **D.** $a^n u(n+2)$

Answer Key

Q2.26.1 A	Q2.25.1 A,C	Q2.25.2 C
Q2.22.1 D	Q2.19.1 A	Q2.14.1 A
Q2.13.1 C	Q2.13.2 B	Q2.12.1 D
Q2.11.1 D	Q2.10.1 C	Q2.09.1 D
Q2.08.1 D	Q2.01.1 D	

UNIT

III

Continuous Time Fourier Series

Topics

1. Introduction to Fourier Series
2. Exponential Fourier Series
3. Predicting the Nature of a Signal from Fourier Coefficients
4. Power Using Fourier Series Coefficients
5. Properties of Exponential Fourier Series
6. Fourier Series of Periodic Waveforms
7. Trigonometric Fourier Series and Symmetry
8. Polar Form, Dirichlet's Conditions and Gibbs Phenomenon

PrepFusion

Unit Snapshot*

Stream: IN	**Questions:** 5	**MCQ:** 5	**MSQ:** 0	**NAT:** 0
1M: 3	**2M:** 2	**Descriptive:** 0	**Avg Weightage** ~2M	**Range:** 1M(2018)–2M(2022)

**See preface for how Avg Weightage and Range are calculated.*

Q3.22.1 MCQ 2022 2M

A periodic function $f(x)$, with period 2, is defined as

$$f(x) = \begin{cases} -1 - x, & -1 \le x < 0, \\ 1 - x, & 0 < x \le 1. \end{cases}$$

The Fourier series of this function contains _____.

(A) Both $\cos(n\pi x)$ and $\sin(n\pi x)$, where $n = 1, 2, 3, \ldots$

(B) Only $\sin(n\pi x)$, where $n = 1, 2, 3, \ldots$

(C) Only $\cos(n\pi x)$, where $n = 1, 2, 3, \ldots$

(D) Only $\cos(2n\pi x)$, where $n = 1, 2, 3, \ldots$

Q3.18.1 MCQ 2018 1M

An ideal square wave with period of 20 ms shown in the figure, is passed through an ideal low pass filter with cut-off frequency 120 Hz. Which of the following is an accurate description of the output?

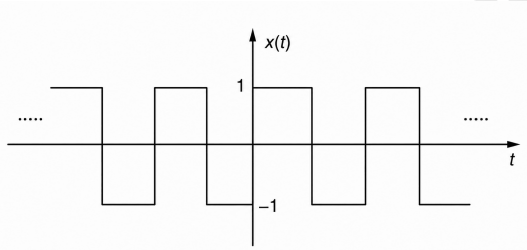


Fig. Q3.18.1

(A) Output is zero.

(B) Output consists of both 50 Hz and 100 Hz frequency components.

(C) Output is a pure sinusoid of frequency 50 Hz.

(D) Output is a square wave of fundamental frequency 50 Hz.

Q3.16.1 MCQ 2016 2M

For the periodic signal $x(t)$ shown below with period $T = 8$ s, the power in the 10th harmonic is

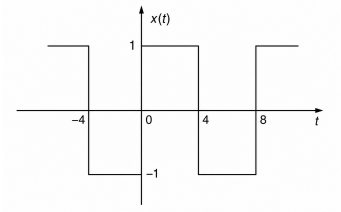


Fig. Q3.16.1

A. 0

B. $\frac{1}{2}\left(\frac{2}{10\pi}\right)^2$

C. $\frac{1}{2}\left(\frac{4}{10\pi}\right)^2$

D. $\frac{1}{2}\left(\frac{4}{5\pi}\right)^2$

Q3.11.1 MCQ 2011 1M

Consider a periodic signal $x(t)$ as shown below.It has a Fourier series representation

$$x(t) = \sum_{k=-\infty}^{\infty} a_k e^{j(2\pi/T)kt}.$$

Which one of the following statement is TRUE?

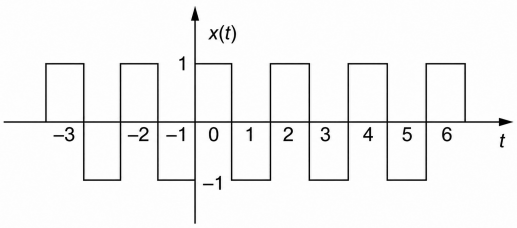


Fig. Q3.11.1

(A) $a_k = 0$, for k odd integer and $T = 2$

(B) $a_k = 0$, for k even integer and $T = 2$

(C) $a_k = 0$, for k even integer and $T = 4$

(D) $a_k = 0$, for k odd integer and $T = 4$

Q3.10.1 MCQ 2010 1M

A periodic signal $f(x)$, shown in the figure, is represented by

$$f(x) = a_0 + \sum_{n=1}^{\infty} (a_n \cos(nx) + b_n \sin(nx)).$$

The value of a_0 is

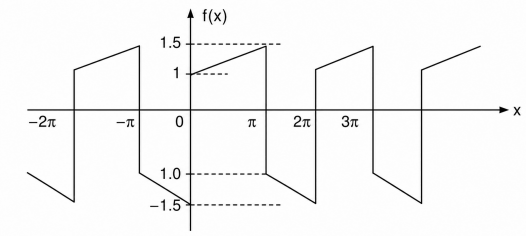


Fig. Q3.10.1

A. 0 **B.** $\frac{\pi}{2}$

C. π **D.** 2π

Answer Key

Q3.22.1 B	Q3.18.1 C	Q3.16.1 A
Q3.11.1 B	Q3.10.1 A	

PrepFusion

UNIT

IV

Continuous Time Fourier Transform

Topics

1. Introduction to Fourier Transform
2. Properties of Fourier Transform
3. Fourier Transform of Standard Signals
4. Duality Property
5. FT of Rectangular, Triangular and Sampling Signals
6. Fourier Transform of Periodic Signals
7. Differentiation, Integration, Multiplication and Convolution Property
8. Parseval's Theorem
9. LTI Systems and Analog Filters
10. Nyquist Rate - Baseband and Bandpass Signals
11. Sampling Theorem and Applications

Unit Snapshot*

Stream: IN	**Questions:** 26	**MCQ:** 20	**MSQ:** 1	**NAT:** 5
1M: 13	**2M:** 13	**Descriptive:** 0	**Avg Weightage** ~3M	**Range:** 1M(2026)–4M(2023)

**See preface for how Avg Weightage and Range are calculated.*

Q4.26.1 MCQ 2026 1M

A square wave signal of frequency 20, kHz is passed through an ideal low-pass filter with cut-off frequency of 21, kHz. The output signal is a _____.

A. 20 kHz sine wave **B.** 20 kHz square wave
C. 20 kHz triangular wave **D.** 20 kHz saw-tooth wave

Q4.25.1 MCQ 2025 2M

Let $f(t)$ and $g(t)$ represent continuous-time real-valued signals. If $h(t)$ denotes cross-correlation between $f(t)$ and $g(-t)$, its continuous-time Fourier transform $H(j\omega)$ equals.

Note: $F(j\omega)$ and $G(j\omega)$ denote the continuous-time Fourier transforms of $f(t)$ and $g(t)$, respectively.

A. $F(j\omega)G(j\omega)$ **B.** $F(-j\omega)G(j\omega)$
C. $F(j\omega)G(-j\omega)$ **D.** $-F(j\omega)G(-j\omega)$

Q4.24.1 NAT 2024 1M

Consider a filter defined by the difference equation

$$y[n] - 0.5\,y[n-2] = ax[n-4]$$

where $x[n]$ and $y[n]$ represent the input and output, respectively. If the magnitude response of the filter at $\omega = \frac{\pi}{2}$ is $\left|H\left(\frac{\pi}{2}\right)\right| = 0.5$,the value of a is _____.(Rounded off to two decimal places.)

Q4.24.2 NAT 2024 1M

The Nyquist sampling frequency for

$$x(t) = 10\sin^2(200\pi t)$$

is _____ Hz (rounded off to nearest integer).

Q4.23.1 MSQ 2023 2M

A Continuous real valued signal $x(t)$ has finite positive energy and $x(t) = 0, \forall\, t < 0$. From the list given below, select **ALL** the signals whose continuous-time Fourier transform is purely imaginary.

A. $x(t) + x(-t)$ **B.** $x(t) - x(-t)$
C. $j\,(x(t) + x(-t))$ **D.** $j\,(x(t) - x(-t))$

Q.23.2 MCQ 2023 2M

The impulse response of an LTI system is

$$h(t) = \delta(t) + 0.5\delta(t-4),$$

where $\delta(t)$ is the continuous-time unit impulse signal. If the input signal

$$x(t) = \cos\left(\frac{7\pi}{4}t\right),$$

the output is

A. $0.5\cos\left(\frac{7\pi}{4}t\right)$ **B.** $1.5\cos\left(\frac{7\pi}{4}t\right)$
C. $0.5\sin\left(\frac{7\pi}{4}t\right)$ **D.** $1.5\sin\left(\frac{7\pi}{4}t\right)$

Q4.22.1 MCQ 2022 2M

A signal $x(t)$ is band-limited between 100 Hz and 200 Hz. A signal $y(t)$ is related to $x(t)$ as follows:

$$y(t) = x(2t-5)$$

The statement that is always true is _____.

(A) $y(t)$ is band-limited between 50 Hz and 100 Hz

(B) $y(t)$ is band-limited between 100 Hz and 200 Hz

(C) $y(t)$ is band-limited between 200 Hz and 400 Hz

(D) $y(t)$ is not band-limited

Q4.20.1 NAT 2020 1M

Consider the signal

$$x(t) = e^{-|t|}.$$

Let

$$X(j\omega) = \int_{-\infty}^{\infty} x(t)e^{-j\omega t}\,dt$$

be the Fourier transform of $x(t)$. The value of $X(j0)$ is _____.

Q4.20.2 NAT 2020 2M

A sinusoid of 10 kHz is sampled at 15 k samples/s. The resulting signal is passed through an ideal low-pass filter (LPF) with cut-off frequency of 25 kHz. The maximum frequency component at the output of the LPF (in kHz) is _____.

Q4.19.1 NAT 2019 2M

The output of a continuous-time system $y(t)$ is related to its input $x(t)$ as

$$y(t) = x(t) + \frac{1}{2}x(t-1).$$

If the Fourier transforms of $x(t)$ and $y(t)$ are $X(\omega)$ and $Y(\omega)$, respectively, and $|X(0)|^2 = 4$, the value of $|Y(0)|^2$ is _____.

Q4.18.1 MCQ 2018 2M

The Fourier transform of a signal $x(t)$, denoted by $X(j\omega)$, is shown in the figure.

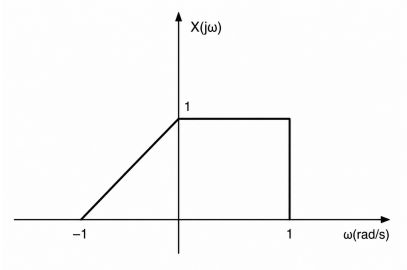


Fig. Q4.18.1

Let

$$y(t) = x(t) + e^{jt}x(t).$$

The value of the Fourier transform of $y(t)$ evaluated at the angular frequency $\omega = 0.5$ rad/s is

A. 0.5 **B.** 1
C. 1.5 **D.** 2.5

Q4.17.1 MCQ 2017 1M

If a continuous-time signal $x(t) = \cos(2\pi t)$ is sampled at 4 Hz, the value of the discrete-time sequence $x[n]$ at $n = 5$ is

A. -0.707 **B.** -1
C. 0 **D.** 1

Q4.15.1 MCQ 2015 1M

The highest frequency present in the signal $x(t)$ is $f_{\max}$. The highest frequency present in the signal $y(t) = x^2(t)$ is

A. $\frac{f_{\max}}{2}$ **B.** $f_{\max}$
C. $2f_{\max}$ **D.** $4f_{\max}$

Q4.12.1 MCQ 2012 2M

The Fourier transform of a signal $h(t)$ is

$$H(j\omega) = (2\cos\omega)\frac{\sin 2\omega}{\omega}.$$

The value of $h(0)$ is

A. $\frac{1}{4}$ **B.** $\frac{1}{2}$
C. 1 **D.** 2

Q4.11.1 MCQ 2011 1M

Consider the signal

$$x(t) = \begin{cases} e^{-t}, & t \geq 0, \\ 0, & t < 0. \end{cases}$$

Let $X(\omega)$ denote the Fourier transform of this signal. The integral

$$\frac{1}{2\pi}\int_{-\infty}^{\infty} X(\omega)\, d\omega$$

is

A. 0 **B.** $\frac{1}{2}$
C. 1 **D.** ∞

Q4.11.2 MCQ 2011 1M

This continuous-time signal

$$x(t) = \cos(100\pi t) + \sin(300\pi t)$$

is sampled at the rate 100 Hz to obtain the signal

$$x_s(t) = \sum_{n=-\infty}^{\infty} x(nT_s)\delta(t - nT_s),$$

where T_s is the sampling period. The signal $x_s(t)$ is passed through an ideal low-pass filter with cut-off frequency 100 Hz. The output of the filter is proportional to

(A) $\cos(100\pi t)$
(B) $\cos(100\pi t) + \sin(100\pi t)$
(C) $\cos(100\pi t) - \sin(100\pi t)$
(D) $\sin(100\pi t)$

Q4.08.1 MCQ 2008 1M

The Fourier transform of

$$x(t) = e^{-at}u(-t),$$

where $u(t)$ is the unit-step function, exists when
(A) it exists for any real value of a
(B) it does not exist for any real value of a
(C) it exists if the real value of a is strictly negative
(D) it exists if the real value of a is strictly positive

Q4.07.1 MCQ 2007 1M

Let $x(t)$ be a continuous-time, real-valued signal band-limited to F Hz. The Nyquist sampling rate (in Hz) for

$$y(t) = x(0.5t) + x(t) - x(2t)$$

is

A. F **B.** $2F$ **C.** $4F$ **D.** $8F$

Q4.06.1 MCQ 2006 2M

The magnitude of Fourier transform $X(\omega)$ of a function $x(t)$ is shown in Figure (a). The magnitude of Fourier transform $Y(\omega)$ of another function $y(t)$ is shown in Figure (b). The phases of $X(\omega)$ and $Y(\omega)$ are zero for all ω. The magnitude and frequency units are identical in both the figures. The function $y(t)$ can be expressed in terms of $x(t)$ as

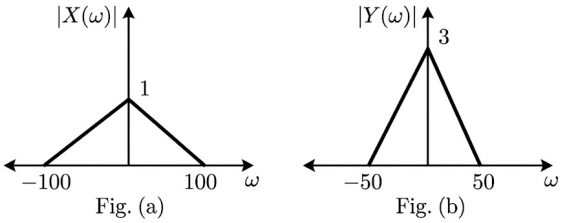


Fig. Q4.06.1

A. $\frac{2}{3}x\left(\frac{t}{2}\right)$ **B.** $\frac{3}{2}x(2t)$

C. $\frac{2}{3}x(2t)$ **D.** $\frac{3}{2}x\left(\frac{t}{2}\right)$

Q4.06.2 MCQ 2006 2M

The Fourier transform of a function $g(t)$ is given by

$$G(\omega) = \frac{\omega^2 + 21}{\omega^2 + 9}.$$

(A) $\delta(t) + 2e^{-3|t|}$

(B) $\cos 3\omega t + 21e^{-3t}$

(C) $\sin 3\omega t + 7\cos \omega t$

(D) $\sin 3\omega t + 21e^{3t}$

Q4.06.3 MCQ 2006 2M

If the waveform shown in the figure corresponds to the second derivative of a given function $f(t)$, then the Fourier transform of $f(t)$ is

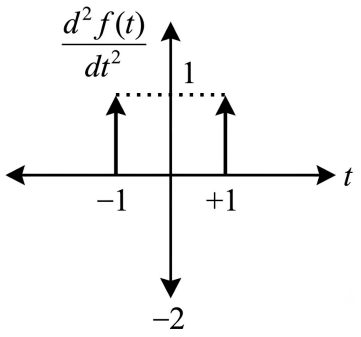


Fig. Q4.06.3

A. $1 + \sin\omega$ **B.** $1 + \cos\omega$

C. $\frac{2(1 - \cos\omega)}{\omega^2}$ **D.** $\frac{2(1 + \cos\omega)}{\omega^2}$

Q4.06.4 MCQ 2006 2M

The spectrum of a band-limited signal after sampling is shown below. The value of the sampling interval is

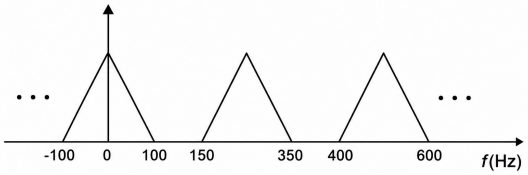


Fig. Q4.06.4

A. 1 ms **B.** 2 ms

C. 4 ms **D.** 8 ms

Q4.06.5 MCQ 2006 2M

A digital measuring instrument employs a sampling rate of 100 samples/second. The sampled input $x[n]$ is averaged using the difference equation

$$y[n] = \frac{x[n] + x[n-1] + x[n-2] + x[n-4]}{4}.$$

For a step input, the maximum time taken for the output to reach the final value after the input transition is

A. 20 ms **B.** 40 ms

C. 80 ms **D.** ∞

Q4.05.1 MCQ 2005 1M

The continuous-time signal

$$x(t) = \frac{1}{a^2 + t^2}$$

has the Fourier transform

$$\frac{\pi}{a} e^{-a|\omega|}.$$

The signal $x(t)\cos(bt)$ has Fourier transform

(A) $\frac{\pi}{2a}\left[e^{-a|\omega-b|} + e^{-a|\omega+b|}\right]$

(B) $\frac{\pi}{2a}\left[e^{-a|\omega|} + e^{-a|\omega|}\right]$

(C) $\frac{\pi}{a} e^{-a|\omega|}\cos(b\omega)$

(D) $\frac{\pi}{2a}\left[e^{-a|\omega-b|} - e^{-a|\omega+b|}\right]$

Q4.04.1 MCQ 2004 1M

A signal

$$x(t) = 5\cos(150\pi t - 60^\circ)$$

is sampled at 200 Hz. The fundamental period of the sampled sequence $x[n]$ is

A. $\frac{1}{200}$ **B.** $\frac{2}{200}$
C. 4 **D.** 8

Q4.03.1 MCQ 2003 1M

A real function $f(t)$ has a Fourier transform $F(\omega)$. The Fourier transform of $f(t) - f(-t)$ is

A. zero **B.** real
C. real and odd **D.** imaginary

Q4.06.1 D	Q4.06.2 A	Q4.06.3 C
Q4.06.4 C	Q4.06.5 B	Q4.05.1 A
Q4.04.1 D	Q4.03.1 D	

Answer Key

Q4.26.1 A	Q4.25.1 A	Q4.24.1 0.70 to 0.80
Q4.24.2 400	Q4.23.1 B,C	Q.23.2 A
Q4.22.1 C	Q4.20.1 2	Q4.20.2 25
Q4.19.1 9	Q4.18.1 C	Q4.17.1 C
Q4.15.1 C	Q4.12.1 C	Q4.11.1 C
Q4.11.2 B	Q4.08.1 C	Q4.07.1 C

UNIT

V

Laplace Transform

Topics

1. Introduction to Laplace Transform
2. Laplace Transform of Standard Signals and Properties
3. Region of Convergence (ROC)
4. Inverse Laplace Transform
5. Interrelation Between EFSC, FT and LT
6. Eigen Values and Eigen Functions in LTI Systems
7. Unilateral Laplace Transform

PrepFusion

Unit Snapshot*

Stream: IN	**Questions:** 17	**MCQ:** 14	**MSQ:** 0	**NAT:** 3
1M: 7	**2M:** 10	**Descriptive:** 0	**Avg Weightage** ~3M	**Range:** 1M(2016)–5M(2022)

**See preface for how Avg Weightage and Range are calculated.*

Q5.26.1 MCQ 2026 2M

Given two signals:

$$x(t) = u(t-2) - u(t-3)$$

and

$$y(t) = e^{-4t}u(t).$$

If

$$z(t) = x(t) * y(t),$$

then $Z(s)$ is _____.

A. $\dfrac{e^{-3s}(e^{s}-1)}{s(s+4)}$ **B.** $\dfrac{e^{-s}(e^{-3s}-1)}{(s+3)(s+4)}$

C. $\dfrac{e^{-3s}(1-e^{s})}{(s+3)(s+4)}$ **D.** $\dfrac{e^{-3s}(1-e^{s})}{s(s+4)}$

Q5.24.1 MCQ 2024 2M

Laplace transform of a signal $x(t)$ is

$$X(s) = \frac{1}{s^2 + 13s + 42}.$$

Let $u(t)$ be the unit step function. Choose the signal $x(t)$ from the following options if the region of convergence is

$$-7 < \Re\{s\} < -6.$$

(A) $-e^{-6t}u(t) - e^{-7t}u(-t)$

(B) $-e^{-6t}u(-t) - e^{-7t}u(t)$

(C) $e^{-6t}u(t) - e^{-7t}u(-t)$

(D) $-e^{-6t}u(-t) - e^{-7t}u(-t)$

Q5.23.1 MCQ 2023 2M

The Laplace transform of the continuous-time signal

$$x(t) = e^{-3t}u(t-5)$$

is _____, where $u(t)$ denotes the continuous-time unit step signal.

A. $\dfrac{e^{-5s}}{s+3}$, $\text{Re}\{s\} > -3$ **B.** $\dfrac{e^{-5(s-3)}}{s-3}$, $\text{Re}\{s\} > 3$

C. $\dfrac{e^{-5(s+3)}}{s+3}$, $\text{Re}\{s\} > -3$ **D.** $\dfrac{e^{-5(s-3)}}{s+3}$, $\text{Re}\{s\} > -3$

Q5.23.2 MCQ 2023 2M

The impulse response of an LTI system is

$$h(t) = \delta(t) + 0.5\,\delta(t-4),$$

where $\delta(t)$ is the continuous-time unit impulse signal. If the input signal is

$$x(t) = \cos\left(\frac{7\pi}{4}t\right),$$

the output is _____.

A. $0.5\cos\left(\dfrac{7\pi}{4}t\right)$ **B.** $1.5\cos\left(\dfrac{7\pi}{4}t\right)$

C. $0.5\sin\left(\dfrac{7\pi}{4}t\right)$ **D.** $1.5\sin\left(\dfrac{7\pi}{4}t\right)$

Q5.22.1 MCQ 2022 1M

The input $x(t)$ to a system is related to its output $y(t)$ as

$$\frac{dy(t)}{dt} + y(t) = 3x(t-3)u(t-3)$$

Here $u(t)$ represents a unit-step function. The transfer function of this system is _____.

A. $\dfrac{e^{-3s}}{s+3}$ **B.** $\dfrac{3e^{-3s}}{s+1}$

C. $\dfrac{3e^{-s/3}}{s+1}$ **D.** $\dfrac{e^{-s/3}}{s+3}$

Q5.22.2 MCQ 2022 2M

The signal

$$x(t) = (t-1)^2u(t-1),$$

where $u(t)$ is the unit-step function, has the Laplace transform $X(s)$.

The value of $X(1)$ is _____.

A. $\dfrac{1}{e}$ **B.** $\dfrac{2}{e}$

C. $2e$ **D.** e^2

Q5.22.3 MCQ 2022 2M

A unity-gain negative-feedback control system has a loop-gain $L(s)$ given by,

$$L(s) = \frac{6}{s(s-5)}.$$

The closed-loop system is

A. Non-causal and unstable **B.** Causal and unstable

C. Non-causal and stable **D.** Causal and stable

Q5.21.1 MCQ 2021 1M

Let $u(t)$ denote the unit step function. The bilateral Laplace transform of the function

$$f(t) = e^{t}u(-t)$$

is _____.

A. $\frac{1}{s-1}$ with real part of $s < 1$ **B.** $\frac{1}{s-1}$ with real part of $s > 1$

C. $\frac{-1}{s-1}$ with real part of $s < 1$ **D.** $\frac{-1}{s-1}$ with real part of $s > 1$

Q5.21.2 NAT 2021 2M

Given

$$y(t) = e^{-3t}u(t) * u(t+3),$$

where $*$ denotes convolution operation. The value of $y(t)$ as $t \to \infty$ is _____ (rounded off to two decimal places).

Q5.17.1 NAT 2017 2M

The Laplace transform of a causal signal $y(t)$ is

$$Y(s) = \frac{s+2}{s+6}.$$

The value of the signal $y(t)$ at $t = 0.1$ s is _____ units.

Q5.17.2 MCQ 2017 1M

A system is described by the following differential equation:

$$\frac{dy(t)}{dt} + 2y(t) = \frac{dx(t)}{dt} + x(t),$$

with

$$x(0) = y(0) = 0,$$

where $x(t)$ and $y(t)$ are the input and output variables, respectively. The transfer function of the inverse system is

A. $\frac{s+1}{s-2}$ **B.** $\frac{s+2}{s+1}$

C. $\frac{s+1}{s+2}$ **D.** $\frac{s-1}{s-2}$

Q5.16.1 NAT 2016 1M

If $X(s)$, the Laplace transform of the signal $x(t)$, is given by

$$X(s) = \frac{s+2}{(s+1)(s+3)^2},$$

then the value of $x(t)$ as $t \to \infty$ is _____.

Q5.13.1 MCQ 2013 2M

The Laplace transform representation of the triangular pulse shown below is

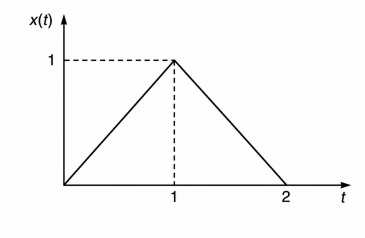


Fig. Q5.13.1

A. $\frac{1}{s^2}\left(1 + e^{-2s}\right)$ **B.** $\frac{1}{s^2}\left(1 - e^{-s} + e^{-2s}\right)$

C. $\frac{1}{s^2}\left(1 - e^{-s} + 2e^{-2s}\right)$ **D.** $\frac{1}{s^2}\left(1 - 2e^{-s} + e^{-2s}\right)$

Q5.12.1 MCQ 2012 1M

The unilateral Laplace transform of $f(t)$ is

$$\frac{1}{s^2+s+1}.$$

The unilateral Laplace transform of $tf(t)$ is

A. $-\frac{s}{(s^2+s+1)^2}$ **B.** $-\frac{2s+1}{(s^2+s+1)^2}$

C. $\frac{s}{(s^2+s+1)^2}$ **D.** $\frac{2s+1}{(s^2+s+1)^2}$

Q5.10.1 MCQ 2010 1M

Let $u(t)$ represent the unit-step function. The Laplace transform of $u(t-\tau)$ is

A. $\frac{1}{s\tau_{s\tau}}$ **B.** $\frac{1}{s-\tau}$

C. $\frac{e^{-s\tau}}{s}$ **D.** $e^{-s\tau}$

Q5.07.1 MCQ 2007 2M

Let the signal $x(t)$ have the Fourier transform $X(\omega)$. Consider the signal

$$y(t) = \frac{d}{dt}\left[x(t-t_d)\right],$$

where t_d is an arbitrary delay. The magnitude of the Fourier transform of $y(t)$ is given by

A. $|X(\omega)|\,|\omega|$ **B.** $|X(\omega)|\,\omega$

C. $\omega^2|X(\omega)|$ **D.** $|\omega||X(\omega)|e^{-j\omega t_d}$

Q5.05.1 MCQ 2005 1M

Identify the transfer function corresponding to an all-pass filter from the following:

A. $\frac{1 - s\tau}{1 + s\tau}$

B. $\frac{1 + s\tau_1}{1 + s\tau_2}$

C. $\frac{1}{1 + s\tau}$

D. $\frac{s\tau}{1 + s\tau}$

Answer Key

Q5.26.1 A	Q5.24.1 B	Q5.23.1 C
Q5.23.2 A	Q5.22.1 B	Q5.22.2 B
Q5.22.3 B	Q5.21.1 C	Q5.21.2 0.33
Q5.17.1 -2.4 to -2.0	Q5.17.2 B	Q5.16.1 0
Q5.13.1 D	Q5.12.1 D	Q5.10.1 C
Q5.07.1 A	Q5.05.1 A	

PrepFusion

UNIT

VI

Z Transform

Topics

1. Introduction to Z-Transform
2. Z-Transform and ROC
3. Inverse Z-Transform
4. Important Properties of Z-Transform
5. Unilateral Z-Transform

PrepFusion

Unit Snapshot*

Stream: IN	**Questions:** 21	**MCQ:** 20	**MSQ:** 0	**NAT:** 1
1M: 10	**2M:** 11	**Descriptive:** 0	**Avg Weightage** ~3M	**Range:** 1M(2017)–4M(2024)

**See preface for how Avg Weightage and Range are calculated.*

Q6.24.1 NAT 2024 2M

Consider a filter defined by the difference equation

$$y[n] - 0.5\,y[n-2] = a\,x[n-4],$$

where $x[n]$ and $y[n]$ represent the input and output, respectively. If the magnitude response of the filter at $\omega = \frac{\pi}{2}$ is $\left|H\left(\frac{\pi}{2}\right)\right| = 0.5$, the value of a is

Q6.24.2 MCQ 2024 2M

Consider a discrete-time sequence

$$x[n] = \begin{cases} (0.2)^n, & 0 \le n \le 7 \\ 0, & \text{otherwise} \end{cases}$$

The region of convergence of $X(z)$, the z-transform of $x[n]$, consists of

A. all values of z except $z = 0.2$

B. all values of z

C. all values of z except $z = 0$

D. all values of z except $z = \infty$

Q6.23.1 MCQ 2023 2M

$F(z) = \frac{1}{1-z}$, when expanded as a power series around $z = 2$, would result in $F(z) = \sum_{k=0}^{\infty} a_k (z-2)^k$, with the region of convergence $|z-2| < 1$. The coefficients a_k, $k \ge 0$, are given by the expression

A. $(-1)^k$ **B.** $(-1)^{k+1}$ **C.** $\left(\frac{1}{2}\right)^k$ **D.** $\left(\frac{-1}{2}\right)^{k+1}$

Q6.18.1 MCQ 2018 2M

Let $y[n] = x[n] * h[n]$, where $*$ denotes convolution and $x[n]$ and $h[n]$ are two discrete time sequences. Given that the z-transform of $y[n]$ is $Y(z) = 2 + 3z^{-1} + z^{-2}$, the z-transform of $p[n] = x[n] * h[n-2]$ is

A. $2 + 3z + z^{-2}$ **B.** $3z + z^{-2}$

C. $2z^2 + 3z + 1$ **D.** $2z^{-2} + 3z^{-3} + z^{-4}$

Q6.17.1 MCQ 2017 1M

Consider two causal discrete-time signals $x_1(n) = \{1, 1\}$ and $x_2(n) = \{1, 2\}$ for $n = 0, 1$. The Z-transform of the convoluted sequence $x_1(n) \otimes x_2(n)$ is

A. $1 + 2z^{-1} + 3z^{-2}$ **B.** $z^2 + 3z + 2$

C. $1 + 3z^{-1} + 2z^{-2}$ **D.** $z^{-2} + 3z^{-3} + 2z^{-4}$

Q6.15.1 MCQ 2015 1M

The Z-transform of $x[n] = \alpha^{|n|}$, $0 < |\alpha| < 1$ is $X(z)$. The region of convergence of $X(z)$ is

A. $|\alpha| < |z| < \frac{1}{|\alpha|}$ **B.** $|z| > \alpha$

C. $|z| > \frac{1}{|\alpha|}$ **D.** $|z| < \min\left[|\alpha|, \frac{1}{|\alpha|}\right]$

Q6.14.1 MCQ 2014 2M

The transfer function of a digital system is given by

$$\frac{b_0}{1 - z^{-1} + a_2 z^{-2}}$$

where a_2 is real. The transfer function is BIBO stable if the value of a_2 is

A. -1.5 **B.** -0.75 **C.** 0.5 **D.** 1.5

Q6.14.2 MCQ 2014 2M

The system function of an LTI system is given by,

$$H(z) = \frac{1 - \frac{1}{3}z^{-1}}{1 - \frac{1}{4}z^{-1}}.$$

The above system can have stable inverse if the region of convergence of $H(z)$ is defined as

A. $|z| < \frac{1}{4}$ **B.** $|z| < \frac{1}{12}$

C. $|z| > \frac{1}{4}$ **D.** $|z| < \frac{1}{3}$

Q6.13.1 MCQ 2013 1M

The discrete-time transfer function $\frac{1 - 2z^{-1}}{1 - 0.5z^{-1}}$ is

A. non-minimum phase and unstable

B. minimum phase and unstable

C. minimum phase and stable

D. non-minimum phase and stable

Q6.12.1 MCQ 2012 2M

If $x[n] = \left(\frac{1}{3}\right)^{|n|} - \left(\frac{1}{2}\right)^{n} u[n]$, then the region of convergence (ROC) of its Z-transform in the z-plane will be

A. $\frac{1}{3} < |z| < 3$ **B.** $\frac{1}{3} < |z| < \frac{1}{2}$

C. $\frac{1}{2} < |z| < 3$ **D.** $\frac{1}{3} < |z| < 2$

Q6.11.1 MCQ 2011 2M

Consider the difference equation

$$y[n] - \frac{1}{3}y[n-1] = x[n]$$

and suppose that $x[n] = \left(\frac{1}{2}\right)^{n} u[n]$. Assuming the conditions of initial rest, the solution of $y[n]$, $n \geq 0$ is

A. $3\left(\frac{1}{3}\right)^{n} - 2\left(\frac{1}{2}\right)^{n}$ **B.** $-2\left(\frac{1}{3}\right)^{n} + 3\left(\frac{1}{2}\right)^{n}$

C. $2\left(\frac{1}{3}\right)^{n} + \frac{1}{3}\left(\frac{1}{2}\right)^{n}$ **D.** $\frac{1}{3}\left(\frac{1}{3}\right)^{n} + \frac{2}{3}\left(\frac{1}{2}\right)^{n}$

Q6.10.1 MCQ 2010 2M

$H(z)$ is a discrete rational transfer function. To ensure that both $H(z)$ and its inverse are stable, its

A. poles must be inside the unit circle and zeros must be outside the unit circle

B. poles and zeros must be inside the unit circle

C. poles and zeros must be outside the unit circle

D. poles must be outside the unit circle and zeros should be inside the unit circle

Q6.08.1 MCQ 2008 1M

The region of convergence of the Z-Transform of the discrete-time signal $x[n] = 2^{n}u[n]$ will be

A. $|z| > 2$ **B.** $|z| < 2$ **C.** $|z| > \frac{1}{2}$ **D.** $|z| < \frac{1}{2}$

Q6.08.2 MCQ 2008 2M

Consider a discrete-time system for which the input $x[n]$ and the output $y[n]$ are related as,

$$y[n] = x[n] - \frac{1}{3}y[n-1].$$

If $y[n] = 0$ for $n < 0$ and $x[n] = \delta[n]$, then $y[n]$ can be expressed in terms of the unit step $u[n]$ as

A. $\left(-\frac{1}{3}\right)^{n} u[n]$ **B.** $\left(\frac{1}{3}\right)^{n} u[n]$

C. $(3)^{n}u[n]$ **D.** $(-3)^{n}u[n]$

Q6.05.1 MCQ 2005 2M

A discrete-time system is shown in the figure. The system function $H(z)$ of the network is given by

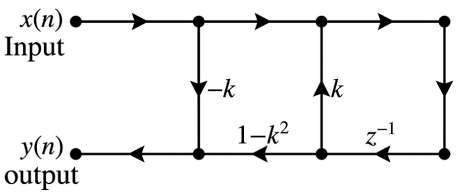


Fig. Q6.05.1

A. $\frac{-k + z^{-1}}{1 - (1-k^2)z^{-1}}$ **B.** $\frac{(1-k^2) + z^{-1}}{1 - kz^{-1}}$

C. $\frac{1 - (1-k^2) + z^{-1}}{1 - kz^{-1}}$ **D.** $\frac{-k + z^{-1}}{1 - kz^{-1}}$

Q6.04.1 MCQ 2004 1M

Given

$$X(z) = \frac{1/2}{1 - az^{-1}} + \frac{1/3}{1 - bz^{-1}}, |a| < 1, |b| < 1$$

with the ROC specified as $|a| < |z| < |b|$, $x[0]$ for the corresponding sequence is given by

A. $\frac{1}{3}$ **B.** $\frac{5}{6}$ **C.** $\frac{1}{2}$ **D.** $\frac{1}{6}$

Q6.04.2 MCQ 2004 1M

A direct form implementation of an LTI system with

$$H(z) = \frac{1}{1 - 0.7z^{-1} + 0.13z^{-2}}$$

is shown in the figure. The value of a_0, a_1 and a_2 are respectively

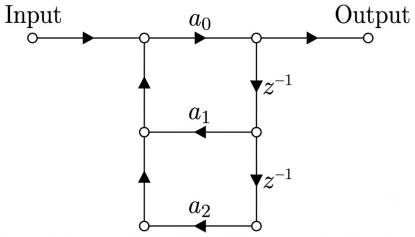


Fig. Q6.04.2

A. 1.0, 0.7 and − 0.13 **B.** −0.13, 0.7 and 1.0

C. 1.0, −0.7 and 0.13 **D.** 0.13, −0.7 and 1.0

Q6.04.3 MCQ 2004 1M

A discrete-time signal, $x[n]$, suffered a distortion modeled by an LTI system with

$$H(z) = (1 - az^{-1})$$

where a is real and $|a| > 1$. The impulse response of a stable system that exactly compensates the magnitude of the distortion is

A. $\left(\frac{1}{a}\right)^n u[n]$ **B.** $-\left(\frac{1}{a}\right)^n u[-n-1]$

C. $a^n u[n]$ **D.** $-a^n u[-n-1]$

Q6.03.1 MCQ 2003 1M

The sequence $x[n]$ whose Z-Transform is $X(z) = e^{1/z}$ is

A. $\frac{1}{n!}u[n]$ **B.** $\frac{1}{n!}u[-n]$

C. $(-1)^n \frac{1}{n!}u[n]$ **D.** $\frac{1}{-(n+1)!}u[-n-1]$

Q6.03.2 MCQ 2003 1M

Given $x = [a, b, c, d]$ as the input, a linear time invariant system produces an output

$$y = [x, x, x, x, \ldots, \langle \text{repeated } N \text{ times} \rangle]$$

The impulse response of the system is

A. $\sum_{i=0}^{N-1} \delta[n - 4i]$

B. $u[n] - u[n - N]$

C. $u[n] - u[n - N - 1]$

D. $\sum_{i=0}^{N-1} \delta[n - i]$

Q6.01.1 MCQ 2001 1M

The impulse response of a discrete LTI system is $a^n u[n]$. Its step response is given by

A. $\sum_{j=0}^{n} a^j$ **B.** $\sum_{j=0}^{\infty} a^j$ **C.** $\frac{1}{1-a}$ **D.** $\sum_{j=-\infty}^{\infty} a^j$

Answer Key

Q6.24.1 0.75	Q6.24.2 C	Q6.23.1 B
Q6.18.1 D	Q6.17.1 C	Q6.15.1 A
Q6.14.1 C	Q6.14.2 C	Q6.13.1 D
Q6.12.1 C	Q6.11.1 B	Q6.10.1 B
Q6.08.1 A	Q6.08.2 A	Q6.05.1 D
Q6.04.1 C	Q6.04.2 A	Q6.04.3 D
Q6.03.1 A	Q6.03.2 A	Q6.01.1 A

UNIT

VII

DTFT, DTFS, DFT & Miscellaneous

Topics

1. DTFT
2. Discrete-Time Fourier Series (DTFS)
3. Circular Convolution
4. DFT
5. Group Delay, Phase Delay and Basics of FFT

PrepFusion

Unit Snapshot*

Stream: IN	**Questions:** 35	**MCQ:** 30	**MSQ:** 0	**NAT:** 5
1M: 9	**2M:** 26	**Descriptive:** 0	**Avg Weightage** ~3M	**Range:** 2M(2025)–5M(2015)

**See preface for how Avg Weightage and Range are calculated.*

Q7.25.1 NAT 2025 2M

Let $X\left(e^{j\omega}\right)$ represent the discrete time Fourier transform of a 4-length sequence $x[n]$, where $x[0] = 1$, $x[1] = 2$, $x[2] = 2$, and $x[3] = 4$. $X\left(e^{j\omega}\right)$ is sampled at $\omega_k = \dfrac{2\pi k}{3}$ to generate a periodic sequence in k with period 3, where k represents an integer. Let $y[n]$ represent another sequence such that its discrete Fourier transform $Y[k]$ is given as $Y[k] = X\left(e^{j\omega_k}\right)$ for $0 \le k \le 2$. The value of $y[0]$ is _____ (in integer).

Q7.24.1 NAT 2024 1M

The number of complex multiplications required for computing a 16-point DFT using the decimation-in-time radix-2 FFT is _____(in integer).

Q7.24.2 NAT 2024 2M

The 4-point DFTs of two sequences $x[n]$ and $y[n]$ are

$$X[k] = [1, -j, 1, j]$$

and

$$Y[k] = [1, 3j, 1, -3j],$$

respectively. Assuming $z[n]$ represents the 4-point circular convolution of $x[n]$ and $y[n]$, the value of

$$z[0] = _____$$

(rounded off to nearest integer).

Note: The DFT of a N-point sequence $x[n]$ is defined as

$$X[k] = \sum_{n=0}^{N-1} x[n]e^{-j\frac{2\pi nk}{N}}.$$

Q7.20.1 NAT 2020 2M

Consider the finite sequence $X = (1, 1, 1)$. The Inverse Discrete Fourier Transform (IDFT) of X is given as $(x(0), x(1), x(2))$. The value of $x(2)$ is _____.

Q7.19.1 NAT 2019 2M

The frequency response of a digital filter $H(\omega)$ has the following characteristics:

Passband: $0.95 \le |H(\omega)| \le 1.05, \quad 0 \le \omega \le 0.3\pi,$

and

Stopband: $0 \le |H(\omega)| \le 0.005, \quad 0.4\pi \le \omega \le \pi,$

where ω is the normalized angular frequency in rad/sample.If the analog upper cut-off frequency for the passband of the above digital filter is 1.2 kHz, then the sampling frequency should be _____ kHz.

Q7.18.1 MCQ 2018 2M

For the sequence

$$x[n] = \{1, -1, 1, -1\}, \qquad n = 0, 1, 2, 3,$$

the DFT is computed as

$$X(k) = \sum_{n=0}^{3} x[n]e^{-j\frac{2\pi}{4}nk},$$

for $k = 0, 1, 2, 3$.The value of k for which $X(k)$ is non-zero is

A. 0 **B.** 1
C. 2 **D.** 3

Q7.17.1 MCQ 2017 2M

Three DFT coefficients, out of the DFT coefficients of a five-point real sequence, are given as

$$X(0) = 4,$$
$$X(1) = 1 - j,$$
$$X(3) = 2 + 2j.$$

The zeroth value of the sequence $x(n)$ is

A. 1 **B.** 2
C. 3 **D.** 4

Q7.15.1 MCQ 2015 2M

The signal

$$x[n] = \frac{\sin\left(\frac{\pi n}{6}\right)}{\pi n}$$

is processed through a linear filter with the impulse response

$$h[n] = \frac{\sin(\omega_c n)}{\pi n},$$

where

$$\omega_c > \frac{\pi}{6}.$$

The output of the filter is

A. $\dfrac{\sin(2\omega_c n)}{\pi n}$ **B.** $\dfrac{\sin(\pi n/3)}{\pi n}$

C. $\left(\dfrac{\sin(\pi n/6)}{\pi n}\right)^2$ **D.** $\dfrac{\sin(\pi n/6)}{\pi n}$

Q7.15.2 MCQ 2015 2M

Consider a low-pass filter module with a pass-band ripple of δ in the gain magnitude. If M such identical modules are cascaded, ignoring the loading effects, the pass-band ripple of the cascaded filter is

A. $1-(1-\delta)^M$ **B.** δ^M
C. $1-\delta^M$ **D.** $(1-\delta)^M$

Q7.15.3 MCQ 2015 1M

Let $3+4j$ be a zero of a fourth-order linear-phase FIR filter. The complex number which is **NOT** a zero of this filter is

A. $3-4j$ **B.** $\frac{3}{25}+\frac{4}{25}j$
C. $\frac{3}{25}-\frac{4}{25}j$ **D.** $\frac{1}{3}-\frac{1}{4}j$

Q7.14.1 MCQ 2014 2M

$X(k)$ is the Discrete Fourier Transform (DFT) of a 6-point real sequence $x(n)$. If $X(0)=9+j0$, $X(2)=2+j2$, $X(3)=3-j0$, $X(5)=1-j1$, then $x(0)$ is

A. 3 **B.** 9
C. 15 **D.** 18

Q7.14.2 MCQ 2014 1M

A discrete-time signal $x[n]$ is obtained by sampling an analog signal at 10 kHz. The signal $x[n]$ is filtered by a system with impulse response

$$h[n]=0.5\,[\delta[n]+\delta[n-1]].$$

The 3 dB cut-off frequency of the filter is

A. 1.25 kHz **B.** 2.50 kHz
C. 4.00 kHz **D.** 5.00 kHz

Q7.11.1 MCQ 2011 1M

Shown below is the pole-zero plot of a digital filter.

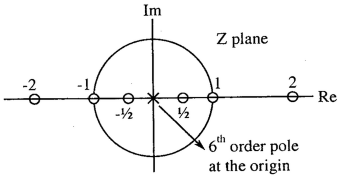


Fig. Q7.11.1

Which one of the following statements is TRUE?

A. This is a low-pass filter. **B.** This is a high-pass filter.
C. This is an IIR filter. **D.** This is an FIR filter.

Q7.11.2 MCQ 2011 1M

A filter is represented by the signal flow graph shown in the figure. Its input is $X(s)$ and output is $Y(s)$. The transfer function of the filter is

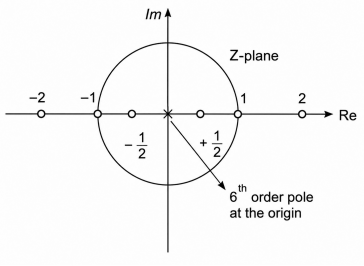


Fig. Q7.11.2

A. $-\frac{1+ks}{s+k}$ **B.** $\frac{1-ks}{s-k}$
C. $-\frac{1-ks}{s+k}$ **D.** $\frac{1-ks}{s+k}$

Q7.10.1 MCQ 2010 2M

The 4-point DFT of a real discrete-time signal $x[n]$ of length 4 is given by $X[k]$, where $k=0,1,2,3$. It is given that

$$X[0]=5, \qquad X[1]=1+j, \qquad X[2]=0.$$

Then $X[3]$ and $x[0]$ respectively are

A. $1-j,\ 1.875$ **B.** $1-j,\ 1.500$
C. $1+j,\ 1.875$ **D.** $0.1-j0.1,\ 1.500$

Q7.10.2 MCQ 2010 1M

A digital filter having the transfer function

$$H(z)=\frac{P_0+P_1z^{-1}+P_3z^{-3}}{1+d_3z^{-3}}$$

is implemented using Direct Form-I and Direct Form-II realizations of an IIR structure. The number of delay units required in Direct Form-I and Direct Form-II realizations are respectively,

A. 6 and 6 **B.** 6 and 3
C. 3 and 3 **D.** 3 and 2

Q7.09.1 MCQ 2009 2M

A linear time-invariant causal system has frequency response given in polar form as

$$H(j\omega)=\frac{1}{\sqrt{1+\omega^2}}\angle\left(-\tan^{-1}\omega\right).$$

For the input

$$x(t) = \sin t,$$

the output is

A. $\frac{1}{\sqrt{2}}\cos t$ **B.** $\frac{1}{\sqrt{2}}\cos\left(t - \frac{\pi}{4}\right)$

C. $\frac{1}{\sqrt{2}}\sin t$ **D.** $\frac{1}{\sqrt{2}}\sin\left(t - \frac{\pi}{4}\right)$

Q7.09.2 MCQ 2009 2M

An analog signal is sampled at 9 kHz. The sequence so obtained is filtered by an FIR filter with transfer function

$$H(z) = 1 - z^{-6}.$$

One of the analog frequencies for which the magnitude response of the filter is zero is

A. 0.75 kHz **B.** 1 kHz

C. 1.5 kHz **D.** 2 kHz

Q7.09.3 MCQ 2009 2M

The transfer function $H(z)$ of a fourth-order linear phase FIR system is given by

$$H(z) = \left(1 + 2z^{-1} + 3z^{-2}\right) G(z).$$

Then $G(z)$ is

A. $3 + 2z^{-1} + z^{-2}$ **B.** $1 + \frac{1}{2}z^{-1} + \frac{1}{3}z^{-2}$

C. $\frac{1}{3} + 2z^{-1} + z^{-2}$ **D.** $1 + 2z + 3z^2$

Q7.06.1 MCQ 2006 2M

A digital filter has the transfer function

$$H(z) = \frac{z^2 + 1}{z^2 + 0.81}.$$

If this filter has to reject a 50 Hz interference from the input, then the sampling frequency for the input signal should be

A. 50 Hz **B.** 100 Hz

C. 150 Hz **D.** 200 Hz

Q7.05.1 MCQ 2005 2M

An analog low-pass filter is needed with the following specifications:

Passband:

$$0.9 \le |H(j\omega)| \le 1, \qquad 0 \le \omega \le 2\pi \times 1000.$$

Stopband:

$$|H(j\omega)| \le 0.2, \qquad |\omega| \ge 2\pi \times 1500.$$

The minimum order N of the Butterworth filter to approximate the above specifications is

A. 6 **B.** 11

C. 12 **D.** 15

Q7.05.2 MCQ 2005 2M

A discrete-time system is shown in the figure. The system function $H(z)$ of the network is given by

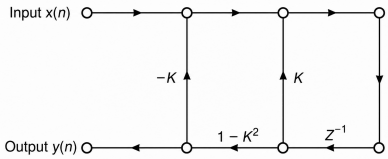


Fig. Q7.05.2

A. $\frac{-k + z^{-1}}{1 - (1 - k^2)z^{-1}}$ **B.** $\frac{(1 - k^2) + z^{-1}}{1 - kz^{-1}}$

C. $\frac{1 - (1 - k^2)z^{-1}}{1 - kz^{-1}}$ **D.** $\frac{-k + z^{-1}}{1 - kz^{-1}}$

Q7.05.3 MCQ 2005 1M

Given the discrete-time sequence

$$x[n] = \{2, 0, -1, -3, 4, 1, -1\},$$

the value of $X(e^{j\pi})$ is

A. 8 **B.** 6π

C. 8π **D.** 6

Q7.04.1 MCQ 2004 2M

A Direct Form implementation of an LTI system with

$$H(z) = \frac{1}{1 - 0.7z^{-1} + 0.13z^{-2}}$$

is shown in the figure. The values of a_0, a_1 and a_2 are respectively

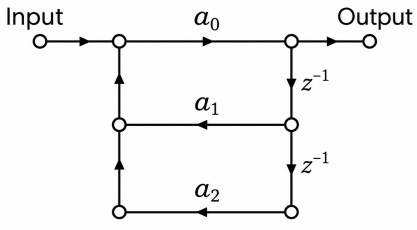


Fig. Q7.04.1

A. 1.0, 0.7, −0.13 **B.** −0.13, 0.7, 1.0
C. 1.0, −0.7, 0.13 **D.** 0.13, −0.7, 1.0

Q7.04.2 MCQ 2004 2M

A low-pass filter has a sampling frequency $f_T = 10$ kHz. The pass-band (f_p) and stop-band (f_s) edge frequencies are 2 kHz and 3 kHz, respectively.The frequency axis is pre-warped before applying bilinear transformation to obtain new angular frequencies, $\bar{r}_p$ and $\bar{r}_s$.The ratio of the warped edge frequencies $\left(\frac{\bar{r}_s}{\bar{r}_p}\right)$ is

A. −1.213 **B.** 1.213
C. 1.575 **D.** 1.894

Q7.04.3 MCQ 2004 2M

A causal analog system has the transfer function

$$H(s) = \frac{a}{s^2 + a^2}.$$

Assuming a sampling time of T seconds, the poles of the transfer function $H(z)$ for an equivalent digital system obtained using the impulse invariance method are at

A. $\left(e^{aT}, e^{-aT}\right)$ **B.** $\left(j\frac{a}{T}, -j\frac{a}{T}\right)$
C. $\left(e^{jaT}, e^{-jaT}\right)$ **D.** $\left(e^{aT/2}, e^{-aT/2}\right)$

Q7.04.4 MCQ 2004 2M

A digital notch filter with a notch frequency of 60 Hz is to be transformed into one operating at a new notch frequency of 120 Hz. The sampling frequency is 400 Hz. A low-pass to low-pass transformation

$$z^{-1} = \frac{1 - \alpha\tilde{z}^{-1}}{\tilde{z}^{-1} - \alpha}$$

is used on the first filter to obtain the new notch filter. Then the value of α should be

A. −0.95 **B.** 0.95
C. −0.46 **D.** 0.46

Q7.04.5 MCQ 2004 1M

If the Fourier transform of $x[n]$ is $X(e^{j\omega})$, then the Fourier transform of $(-1)^n x[n]$ is

A. $(-j)^n X(e^{j\omega})$ **B.** $(-1)^{\omega} X(e^{j\omega})$
C. $X\left(e^{j(\omega-\pi)}\right)$ **D.** $\frac{dX(e^{j\omega})}{d\omega}$

Q7.03.1 MCQ 2003 2M

A linear phase FIR system with real impulse response has a zero at

$$z = \frac{1}{2}e^{j\pi/4}.$$

The largest set of remaining zeros that can be obtained from the above information is

(A) $\frac{1}{2}e^{j\pi/2}, \frac{1}{2}e^{j3\pi/4}, \frac{1}{2}e^{-j\pi/2}$

(B) $2e^{-j\pi/4}, 2e^{-j3\pi/4}, \frac{1}{2}e^{-j\pi/2}$

(C) $2e^{-j\pi/4}, 2e^{j\pi/4}, \frac{1}{2}e^{-j\pi/4}$

(D) $2e^{j\pi/4}, \frac{1}{2}e^{-j\pi/4}$

Q7.03.2 MCQ 2003 2M

Direct-form implementation of the FIR structure shown in the below figure in lattice form. The coefficients K_1 and K_2 are respectively

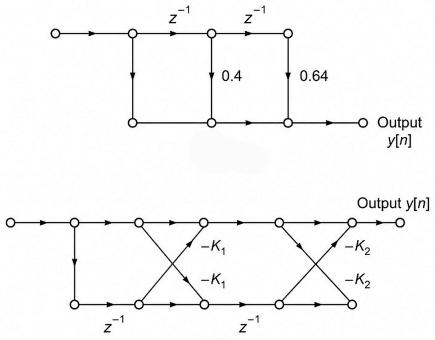


Fig. Q7.03.2

A. 0.640, −0.244 **B.** 0.640, 0.244
C. 0.244, 0.640 **D.** −0.244, −0.640

Q7.03.3 MCQ 2003 2M

A discrete-time system with input $x[n]$ and output $y[n]$ relation

$$y[n] = x[n] - 2x[n-1] + x[n-2]$$

is a good approximation to a

(A) High pass filter

(B) Band-stop filter blocking $\frac{\pi}{8} \le |\omega| \le \frac{\pi}{4}$

(C) Low pass filter

(D) Band-pass filter passing $\frac{\pi}{8} \le |\omega| \le \frac{\pi}{4}$

Q7.03.4 MCQ 2003 2M

Bilinear transformation avoids the problems of aliasing connected with the use of impulse invariance through

(A) Mapping the entire imaginary axis of the s-plane on the unit circle in the z-plane

(B) Pre-filtering the input signal to impose band-limitedness

(C) Mapping zeros of the left half of the s-plane inside the unit circle in the z-plane

(D) Up-sampling the input signal so that the bandwidth is reduced

Q7.03.5 MCQ 2003 2M

A linear time-invariant system with system function

$$H(z) = 1 + z^{-1} + z^{-2}$$

is given an input signal sampled at 18 kHz. The frequency of the analog sinusoid which cannot pass through the system is

A. $\frac{12}{\pi}$ kHz **B.** 6 kHz

C. 12 kHz **D.** $\frac{6}{\pi}$ kHz

Q7.02.1 MCQ 2002 1M

The bilinear transformation

$$\omega = \frac{z-1}{z+1}$$

(A) Maps the inside of the unit circle in the z-plane to the left half of the ω-plane

(B) Maps the outside of the unit circle in the z-plane to the left half of the ω-plane

(C) Maps the inside of the unit circle in the z-plane to the right half of the ω-plane

(D) Maps the outside of the unit circle in the z-plane to the right half of the ω-plane

Q7.96.1 MCQ 1996 2M

A linear phase low-pass filter with a cut-off frequency of 1500 Hz shifts a 50 Hz signal by 3 ms. It will shift a signal of 150 Hz by

A. 3 ms **B.** 9 ms

C. 1 ms **D.** 30 ms

Answer Key

Q7.25.1 5	Q7.24.1 32	Q7.24.2 2
Q7.20.1 0	Q7.19.1 8	Q7.18.1 C
Q7.17.1 B	Q7.15.1 D	Q7.15.2 A
Q7.15.3 D	Q7.14.1 A	Q7.14.2 B
Q7.11.1 D	Q7.11.2 D	Q7.10.1 A
Q7.10.2 B	Q7.09.1 D	Q7.09.2 C
Q7.09.3 A	Q7.06.1 D	Q7.05.1 A
Q7.05.2 D	Q7.05.3 D	Q7.04.1 A
Q7.04.2 MTA(1.349)	Q7.04.3 C	Q7.04.4 C
Q7.04.5 C	Q7.03.1 C	Q7.03.2 D
Q7.03.3 D	Q7.03.4 B	Q7.03.5 B
Q7.02.1 A	Q7.96.1 A	

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